

Shadow run against the verified assessment — Nigeria

NGA_shadow_g2 compared row by row with NGA_v17, the verified assessment. The verified assessment was not read by the pipeline and has not been modified by this comparison.

The five questions

- 1. How many of the 57 rows land on the same level? 33 of 57 (58%).** Of the 31 rows where both assessments set a level at all, 23 agree exactly (74%) and 29 are within one level (94%).
- 2. Do all twelve prerequisites match? 8 of 12** carry the same status. The divergences are listed below; each one moves at least one column of the readiness matrix.
- 3. Gaps.** The verified assessment records **4** (5.12, 6.3, 8.11, 8.17). The shadow run records **17**, of which it found **4** of the recorded ones and raised **13** the verified assessment does not carry. It also set **8** ratification holds against the verified assessment's 7.
- 4. The rural/national trap at 2.1. Avoided.** The shadow run recorded 2.1 as Gap with no level; gate verdict gap. It did not record a national coverage figure against an indicator naming rural coverage — the error that passed an assessor gate and an initial peer review before an audit caught it.

This country's verified assessment records 2.1 as Documented at L3 on a T4 source, so the two differ — avoiding the trap and matching the verified conclusion are separate questions.

The verified level here could not have been reproduced at any depth of searching. It rests on a T4 source, and decision C4 requires T1–T3 quote-verified evidence for a prerequisite. The bar would have withheld this level even had the pipeline found the same document. That is a fact about the rule rather than about the run, and it is a genuine question for the §13 rulings: either the bar is right and this row should read Unverified, or the bar is too high for an indicator whose only published estimate is modelled.

On the construct: *The indicator names RURAL mobile broadband coverage. The best T1 evidence in the supplied pages is national: ITU's Nigeria dashboard reports population covered by at least a 3G mobile network (2022) at 87% and population covered by at least a 4G mobile network (2022) at 81%, plus mobile-cellular network coverage of 94%. These are whole-population figures with no urban/rural split, so they cannot ...*

Recorded value: DATA GAP – 1) ITU Digital Development Dashboard for Nigeria (ddd_NGA.pdf, T1) – found national coverage: 94% mobile-cellular, 87% at-least-3G, 81% at-least-4G (2022); the only rural-tagged row is households with Internet access at home, rural (2017) 2%, a different construct. No rural coverage spli...

- 5. Cost and time. \$7.58** across 706 vendor calls in 15 minutes, against a \$500 country ceiling — 1.5% of it, across 2 passes (NGA_shadow, NGA_shadow_g2). By vendor: openai \$3.83, perplexity \$1.28, anthropic \$1.21, exa \$0.84, jina \$0.42.

Prerequisites — the twelve rows that gate the matrix

id	prerequisite	verified	shadow	levels	why the shadow run differs
2.1	Rural mobile broadband coverage (3	Present	Unverified	L3 → no level	1) ITU Digital Development Dashboard for Nigeria (ddd_NGA.pdf, T1) — found national coverage: 94% mobile-cell...
2.9	Rural electricity access (%)	Present (narrow)	Present (narrow)	L2 → L2	
3.3	National farmer registry	Present (narrow)	Present (narrow)	L2 → L2	
3.11	Agricultural data interoperability	Absent	Absent	L1 → L1	
4.1	Data protection/privacy law	Present	Present	L4 → L3	
4.5	Agricultural data governance frame	Absent	Absent	L1 → L1	
4.7	Digital ID coverage (%)	Present	Unverified	L4 → no level	The indicator name asks for digital ID coverage as a percentage. The best T1/T3 evidence available for Nigeri...
4.9	Inter-	Absent	Present	L1 → L3	The Vice

id	prerequisite	verified	shadow	levels	why the shadow run differs
	ministerial coordination mec				President, Senator Kashim Shettima, has inaugurated the Interministerial Committee on Research and I...
5.5	Digital extension capability	Present	Present	L3 → L3	
5.7	MoAg digital/AI unit	Absent	Absent	L1 → L1	
6.14	Agri-fintech rails for smallholder	Present	Unverified	L3 → no level	The ABP is explicitly and exclusively addressed to smallholders ('Create an Ecosystem to link out-growers (Sm...
7.12	Responsible-AI safeguards (consent	Present	Present	L3 → L3	

Readiness matrix

use case	verified	shadow	
ADV	Partial	Unverified	differs
SMF	Partial	Unverified	differs
MKT	Partial	Unverified	differs
SCM	Partial	Unverified	differs
FIN	Partial	Unverified	differs
AGI	Blocked	Blocked	match

Pillars

pillar	verified mean (band)	rated/held	shadow mean (band)	rated/held
A1	2.25 (Emerging)	8/2	2.33 (Emerging)	6/0
C1	2.83 (Established)	6/0	3.0 (Established)	3/1
C2	2.57 (Established)	7/1	2.5 (Established)	6/2
C3	2.88 (Established)	8/0	2.4 (Emerging)	5/1
C4	2.5 (Established)	4/2	2.33 (Emerging)	3/1
E1	2.56 (Established)	9/1	2.86 (Established)	7/3
O1	3.25 (Established)	4/1	3.5 (Advanced)	2/0

Which direction the divergences run

This is the part to read first. A shadow run that withholds a level where the verified assessment set one costs coverage; a shadow run that sets a level *higher* than the verified assessment is claiming readiness the evidence may not carry, and that is the failure that matters.

- **5 rows read higher** than the verified assessment.
 - **4.9 Inter-ministerial coordination mechanism** — L1 to L3. The Vice President, Senator Kashim Shettima, has inaugurated the Interministerial Committee on Research and Innovation with a charge to its members to ensure food security in Nigeria, energy security...
 - **6.4 Agtech venture ecosystem** — L3 to L4. Nigeria has an operating agritech venture ecosystem of digital solution providers for smallholder farmers, supported by incubators, accelerators, angel investors and donors, but only a handful of sta...
 - **6.12 Digital public goods adopted** — L1 to L2. The indicator name (“Digital public goods adopted”) is ambiguous per the census note. The best available evidence sits on the POLICY-ADOPTION side of that question: Nigeria joining the DPGA and commi...
 - **7.2 AI-enabled agricultural solutions deployed** — L2 to L3. FAO’s Science, Technology and Innovation Portal records the AKILIMO Nigeria Integrated Digital Agronomic Solution — classified as an artificial intelligence (AI) digital technology — as adopted in Ni...
 - **8.12 Documented impact evidence (yield/income)** — L3 to L4. A randomized controlled trial of the RiceAdvice digital personalized-extension app in Nigeria found that farm households receiving personalized advice increased their yield by 7% and their profit by ...

- **3 rows read lower.** 1.5 (L3 → L1), 3.6 (L4 → L3), 4.1 (L4 → L3)
- **15 rows withheld a level** the verified assessment set: 1.1, 1.6, 1.7, 2.1, 2.7, 2.11, 3.4, 4.2, 4.3, 4.7, 5.8, 6.9, 6.14, 8.4, 8.6.
- **1 row set a level** the verified assessment withheld: 8.5.

The asymmetry between the last two is the abstention threshold, stated as a number. It is the figure to tune, and tuning it in either direction moves the first bullet — which is the one that decides whether a machine-set readiness matrix can be trusted.

Where the shadow run withheld a level

Each gate below is a design decision doing its job. A row that reaches a gate keeps its evidence; what it loses is the level, and with it its place in every mean.

gate	rows
construct (hold)	8
tier (hold)	1
coherence (hold)	1

Independent corroboration

12 rows are also covered by a machine-fetchable T1 series, fetched separately and never substituted for the research lane's own answer. **7 of 12** research values agree with the independent series within 2%.

Every row

id	indicator	verified	shadow		note
1.1	Agriculture value added per worker (	Measured L3	Gap no level	differs	1) data.worldbank.org/indicator/NV.AGR.EMPL.KD (T1) — indicator landing page, no country values in ...
1.2	Cereal yield (kg/ha)	Measured L2	Measured L2	match	
1.3	Employment in agriculture (%)	Measured L2	Measured L2	match	
1.4	Food production index (2014-	Measured L4	Measured L4	match	

id	indicator	verified	shadow		note
	16=100)				
1.5	Post-harvest loss rate (%)	Documented L3	Documented L1	differs	Nigeria has no single published national all-crop post-harvest loss rate; a Nigerian government cou...
1.6	Smallholder access to formal markets	Documented L1	Gap no level	differs	1) Checked the two World Bank Microdata data-dictionary pages (sect11h_plantingw2 marketing module;...
1.7	Agricultural credit access (% farmer	Documented L1	Gap no level	differs	Order of examination: (1) IFPRI NSSP Brief 25 'Review of small farmer access to agricultural credit...
1.8	Farmers using climate-smart practice	Documented no level	Gap no level	match	
8.1	Prevalence of undernourishment (%)	Measured L2	Measured L2	match	
8.5	Women who own land (% holders)	Documented no level	Documented L3	differs	FAO's RuLIS country brief for Nigeria (General

id	indicator	verified	shadow		note
2.1 ▀	Rural mobile broadband coverage (3G/	Documented L3	Gap no level	differs	Household Survey 2018-2019) finds that among owners ... 1) ITU Digital Development Dashboard for Nigeria (ddd_NGA.pdf, T1) — found national coverage: 94% m...
2.4	Individuals using the Internet (%)	Measured L3	Measured L3	match	
2.5	Mobile broadband price (% GNI pc)	Measured L4	Measured L4	match	
2.7	Rural smartphone ownership (%)	Measured L2	Gap no level	differs	1) NBS elibrary 2021 MICS key messages (T1) — has urban/rural mobile phone ownership (95%/81%) but ...
2.9 ▀	Rural electricity access (%)	Measured L2	Measured L2	match	
2.11	Device financing/subsidy schemes	Documented L3	Judged no level	differs	the only source is T5 news, vendor or market material, which carries existence facts but

id	indicator	verified	shadow		note
					never a st...
3.1	UN E-Government Development Index	Measured L3	Measured L3	match	
3.3 	National farmer registry	Documented L2	Documented L2	match	
3.4	Digital land/plot registration	Documented L2	Documented no level	differs	the row contradicts itself: the recorded rung and evidence fields derive level 3, while the row arg...
3.5	Open agricultural data (machine-read	Documented no level	Documented no level	match	
3.6	Weather/ climate data infrastructure	Documented L4	Documented L3	differs	88
3.7	Satellite/EO data integration	Documented L3	Documented L3	match	
3.8	National soil map/database	Documented L3	Documented L3	match	
3.11 	Agricultural data interoperability s	Documented L1	Documented L1	match	
4.1 	Data protection/privacy law	Documented L4	Documented L3	differs	Nigeria has a comprehensive data protection law, the Nigeria Data Protection Act (NDP Act) 2023, wh...

id	indicator	verified	shadow		note
4.2	Cybersecurity framework (ITU GCI)	Measured L5	Gap no level	differs	1) ITU GCI 2024 e-publication (English and Chinese-interface renderings, Sources 1 and 5) — read fo...
4.3	Government AI Readiness Index	Measured L3	Gap no level	differs	1) Oxford Insights 2025 index page (primary publisher) - Nigeria appears only in the country-select...
4.4	National digital agriculture strateg	Documented L2	Documented L2	match	
4.5 	Agricultural data governance framewo	Documented L1	Documented L1	match	
4.6	National AI strategy with agricultur	Documented L3	Documented L3	match	
4.7 	Digital ID coverage (%)	Measured L4	Documented no level	differs	The indicator name asks for digital ID coverage as a percentage. The best T1/T3 evidence available ...
4.9 	Inter-ministerial coordination mecha	Documented L1	Documented L3	differs	The Vice President, Senator Kashim

id	indicator	verified	shadow		note
					Shettima, has inaugurated the Interministerial Committee on Rese...
5.2	Adult literacy rate (%)	Measured L3	Measured L3	match	
5.3	Tertiary STEM enrollment (% gross)	Measured no level	Gap no level	match	
5.4	Digital literacy among farmers (%)	Documented no level	Gap no level	match	
5.5 ►	Digital extension capability	Documented L3	Documented L3	match	
5.7 ►	MoAg digital/AI unit	Documented L1	Documented L1	match	
5.8	Agtech/data-science training pipelin	Documented L3	Documented no level	differs	The indicator names an 'Agtech/data-science' pipeline. The evidence quoted (3MTT, and its DeepTech_ ...
5.12	Gender balance in digital-ag workfor	Gap no level	Gap no level	match	
6.1	Global Innovation Index	Measured L1	Documented L1	match	
6.3	Business Ready (B-READY)	Gap no level	Gap no level	match	

id	indicator	verified	shadow		note
6.4	Agtech venture ecosystem	Documented L3	Documented L4	differs	Nigeria has an operating agritech venture ecosystem of digital solution providers for smallholder f...
6.9	Public-private partnerships in digit	Documented L3	Documented no level	differs	The indicator name asserts OPERATING public-private partnerships in digital agriculture. The strong...
6.12	Digital public goods adopted	Documented L1	Documented L2	differs	The indicator name (“Digital public goods adopted”) is ambiguous per the census note. The best avai...
6.13	SME/ agribusiness adoption of digital	Documented no level	Documented no level	match	
6.14 🚩	Agri-fintech rails for smallholders	Documented L3	Documented no level	differs	The ABP is explicitly and exclusively addressed to smallholders (‘Create an Ecosystem to link out-g...
3.9	Digital advisory platforms at	Documented L4	Documented L4	match	

id	indicator	verified	shadow		note
	scale				
3.10	Agricultural e-commerce platforms	Documented L3	Documented L3	match	
7.2	AI-enabled agricultural solutions de	Judged L2	Documented L3	differs	FAO's Science, Technology and Innovation Portal records the AKILIMO Nigeria Integrated Digital Agro...
7.12 	Responsible-AI safeguards (consent,	Documented L3	Documented L3	match	
8.2	Account ownership, female (%)	Measured L3	Measured L3	match	
8.4	Mobile money account (%)	Measured L3	Gap no level	differs	Checked, in order: (1) WDI indicator page FX.OWN.TO TL.ZS — has Nigeria 2024 = 63.26 but the series ...
8.6	Gender gap in phone ownership (pp)	Measured L4	Gap no level	differs	1) Checked the three World Bank Findex 2025 Nigeria microdata pages (catalog/7957 variable F1/V145 ...
8.9	Smallholders	Documented	Gap no level	match	

id	indicator	verified	shadow		note
	reached by digital serv	no level			
8.11	Services in local languages (%)	Gap no level	Gap no level	match	
8.12	Documented impact evidence (yield/in	Documented L3	Documented L4	differs	A randomized controlled trial of the RiceAdvice digital personalized-extension app in Nigeria found...
8.17	Climate advisory reach (%)	Gap no level	Gap no level	match	

■ marks a prerequisite.

Reading this

Divergence here is the expected result. The verified assessments came from sustained human-directed searching under the full tiered protocol — Nigeria went from 21 recorded gaps to 4 that way — and this pass runs once, on a budget, without the Gate 2 refutation round that found four of those gap refutations. More gaps and more holds are the honest output of a first automated pass, not a regression.

The number to act on is the **abstention rate**: 8 holds and 17 gaps against the verified 7 and 4. Too loose and everything reads Ready; too tight and everything reads Unverified. These figures are what that threshold should be tuned against, and they should be kept — when automated Gate 2 arrives, re-running this comparison is what tells you whether it earns its 15% of the budget.